

# FEM for beam

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June 2026

## 1 Introduction

This project aims to analyze the DOFs of a framework contributed by beams under different boundary conditions and loads. The project is developed in two stages. First, a single Euler–Bernoulli beam is used as a verification example, because its boundary conditions, loading cases and analytic solutions are well understood. This allows us to test the implementation of shape functions, stiffness and mass matrices, load vectors, boundary conditions, and the solution methods. After this verification step, the same finite element structure is extended to a framework consisting of several connected beams.

## 2 Verification on a Single Cantilever Beam

### 2.1 Static simulation

This section presents the static analysis of a single Bernoulli beam subjected to three different conditions. Two independent approaches are used:

- the analytical solution of the static bending equation,
- the finite element method (FEM) solution based extended matrix method.

Comparing these two methods allows us to verify the correctness of the numerical implementation and understand the behavior of the beam under static loading.

We consider a prismatic Bernoulli beam of length  $L$ , Young’s modulus  $E$ , and second moment of area  $I$ . The beam is clamped at the left end and subjected to a constant load density  $q(x) = q_0$  along its entire length. The governing static bending equation is:

$$(EIw'')''(x) = q(x)$$

Boundary conditions for a left-clamped beam are:

- $w(0) = 0$  (zero displacement)
- $w'(0) = 0$  (zero rotation)

### 2.1.1 Analytical solution

Taking a simple cantilever as an example, the original DOF  $u \in R^{2n}$ , where  $n$  is the number of nodes.

$$u = [w_1, w'_1, w_2, w'_2, \dots, w_n, w'_n]^T$$

For a cantilever beam, we fix  $u_1 = 0, u_2 = 0$ , so that

$$u_{free} = [u_3, u_4, \dots, u_{2n}]^T$$

And solve the reduced system:

$$S_{ff}u_f = f_f$$

### 2.1.2 Extended matrix method

Unlike the analytical solution above, which modifies the DOF directly, in the extended matrix method, all the DOF will be saved. The constraints will be applied explicitly by **Lagrange multipliers**. The matrix equation is as follows:

$$\begin{bmatrix} S & C \\ C^T & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda \end{bmatrix} = \begin{bmatrix} f \\ a \end{bmatrix}$$

where:

- $S$ : stiffness matrix;
- $C^T u = a$ : Dirichlet boundary constraints;
- $\lambda$ : constraints reactions;
- $a$ : clamped boundary.

### 2.1.3 Comparison between two methods

To verify the correctness of the static solver, we compared the numerical results obtained from the extended matrix method with analytic solutions for three standard loading cases of a cantilever beam. Three loading cases were tested:

- a constant distributed load ( $q = q_0$ ),
- an end-point force ( $Q_L = P$ ),
- an end-point bending moment ( $M_L = M_0$ ).

The analytic results are plotted and shown as Fig.1.

The FEM results are plotted and shown as Fig.2.

Table1 compares the numerical accuracy and computational cost of the extended matrix FEM solver with the corresponding analytic solutions.

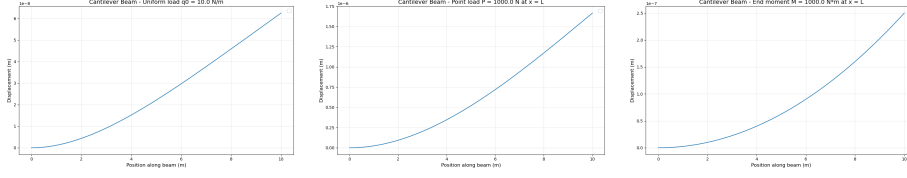


Figure 1: Analytic Solution of Static Simulation

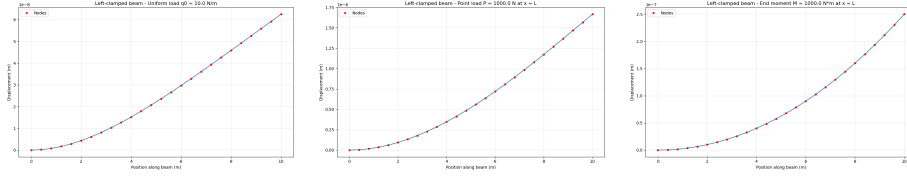


Figure 2: FEM Solution of Static Simulation

The results are summarized in Table 1. For all three cases, the absolute difference between the FEM solution and the analytic solution is on the order of  $10^{-18}$  to  $10^{-19}$ , m. This indicates that the extended matrix method reproduces the analytic solution with very high accuracy for the tested static beam problems.

Overall, the comparison shows that the extended matrix FEM method achieves nearly identical results to the analytic solution while being computationally efficient. The very small errors validate the correctness of the implementation, and the computation time results support the use of the FEM formulation as the basis for later dynamic simulations and framework extensions, where analytic solutions are no longer easily available.

## 2.2 Dynamic simulation with Newmark

### 2.2.1 Description

In this experiment, a cantilever beam model is considered, where the left end is fixed and the right end is free. A constant point load is applied at the free end, and the objective is to analyze the transient dynamic response of the beam under this loading condition. The governing equation of motion is given by

$$M\ddot{u}(t) + D\dot{u}(t) + Ku(t) = f$$

and since damping is neglected in this problem, it reduces to

$$M\ddot{u}(t) + Ku(t) = f$$

The initial conditions are chosen as  $u(0) = 0$  and  $\dot{u}(0) = 0$ , which means that the system starts from an undeformed and stationary state, and the constant load is suddenly applied at time  $t = 0$ .

| Load Case    | Absolute diff(m) | Relative diff | FEM time(s) | Analytic time(s) |
|--------------|------------------|---------------|-------------|------------------|
| Uniform load | 4.84e-19         | 7.74e-12      | 1.65e-02    | 2.95e-01         |
| Point load   | 1.20e-17         | 7.18e-12      | 4.77e-04    | 2.06e-01         |
| End moment   | 1.60e-18         | 6.40e-12      | 3.77e-04    | 2.04e-01         |

Table 1: Comparison between Static Methods

### 2.2.2 Numerical method and algorithmic implementation

The dynamic response is computed using the Newmark method, which is a widely used time integration scheme for second-order differential equations. The method approximates displacement and velocity within each time step and introduces two parameters,  $\beta$  and  $\gamma$ , to control stability and accuracy. For a time step from  $t_n$  to  $t_{n+1}$ , the displacement and velocity are updated based on Taylor's Expansion:

$$u_{n+1} = u_n + \Delta t \dot{u}_n + \Delta t^2 \left[ \left( \frac{1}{2} - \beta \right) \ddot{u}_n + \beta \ddot{u}_{n+1} \right]$$

and

$$\dot{u}_{n+1} = \dot{u}_n + \Delta t ((1 - \gamma) \ddot{u}_n + \gamma \ddot{u}_{n+1}).$$

We can construct the following mediate variables:

$$u_n^* = u_n + \Delta t \dot{u}_j + \left( \frac{1}{2} - \beta \right) \ddot{u}_n$$

$$\dot{u}_n^* = \dot{u}_n + \Delta t (1 - \gamma) \ddot{u}_n$$

By substituting these expressions into the governing equation, a linear system for  $\ddot{u}_{n+1}$  is obtained in the form

$$M_{\text{eff}} \ddot{u}_{n+1} = f_{\text{eff}},$$

where the effective stiffness matrix is

$$M_{\text{eff}} = M + \gamma \Delta t D + \beta \Delta t^2 S,$$

and the effective load vector incorporates contributions from the previous time step as

$$f_{\text{eff}} = f_{n+1} - D \dot{u}_n^* - S u_n^*.$$

At each time step, the procedure consists of constructing the effective system, solving for the new displacement, and then updating acceleration and velocity accordingly. A common and robust choice of parameters is  $\beta = \frac{1}{4}$  and  $\gamma = \frac{1}{2}$ , corresponding to the average acceleration method. This choice ensures unconditional stability and second-order accuracy in time. From a mathematical perspective, the Newmark method can be interpreted as a weighted integration of acceleration and is closely related to a generalized trapezoidal rule, providing stable and reliable results for undamped linear systems.

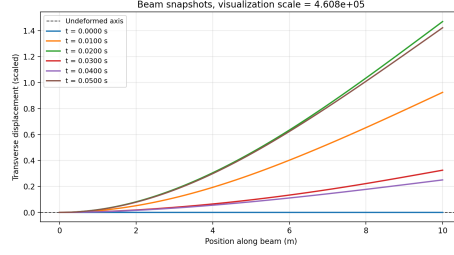


Figure 3: Snapshots of cantilever under point load

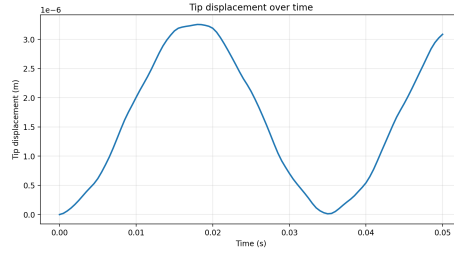


Figure 4: Displacement of free end under point load

### 2.2.3 Numerical results analysis

The numerical results are illustrated using two types of visualizations: deformation snapshots and the displacement history track at the free end.

The snapshots, shown in Fig.15, present the deformation of the beam at different time steps with an appropriate scaling factor applied to enhance visibility. The deformation pattern clearly reflects the boundary conditions: the left end remains fixed, while the free end experiences the largest displacement. The sequence of snapshots also shows how the beam evolves over time, first moving downward under the applied load, then overshooting the equilibrium position, and subsequently reversing direction as part of its oscillatory motion.

In addition to the snapshots, the displacement at the free end is plotted as a function of time in Fig.4. This curve provides a clearer quantitative view of the dynamic response. Since the load is applied suddenly at the initial time, the system is dynamically excited and begins to vibrate. In the absence of damping, the displacement does not converge to a constant value but instead oscillates periodically around the static equilibrium displacement. The amplitude and frequency of this oscillation are consistent with the natural vibration characteristics of the beam, confirming that the system behaves as an undamped oscillator subjected to a constant external load.

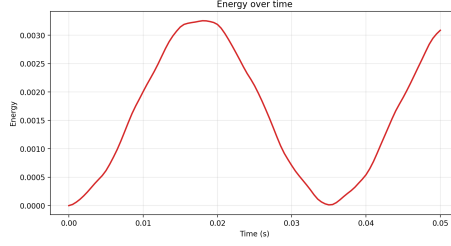


Figure 5: Energy wave of beam

### 2.2.4 Energy behavior

The behavior of energy was also studied, as shown in Fig.5. The mechanical energy of the system consists of kinetic and potential components, defined respectively as

$$E_{\text{kin}} = \frac{1}{2} \dot{u}^T M \dot{u}, \quad E_{\text{pot}} = \frac{1}{2} u^T K u.$$

For a free vibration system without external forces, the total mechanical energy would be conserved. However, in the present case, a constant external force is continuously applied, and therefore the mechanical energy is not conserved.

The variation of energy can be understood from the rate equation

$$\frac{dE_{\text{mech}}}{dt} = \dot{u}^T f,$$

which shows that the external force is the only source of energy input in the absence of damping. When the velocity is aligned with the direction of the applied force, the system gains energy, whereas when the velocity is opposite to the force, the energy decreases. This leads to an oscillatory pattern in the energy curve over time.

A conserved quantity can be obtained by including the potential associated with the external force, leading to

$$E_{\text{total}} = \frac{1}{2} \dot{u}^T M \dot{u} + \frac{1}{2} u^T K u - f^T u.$$

This expression represents the total energy of the system, including the contribution from the external load, and remains constant in the absence of damping.

## 2.3 Free vibration with eigenvalue method

### 2.3.1 Objective

In addition to the Newmark time-integration scheme, the free vibration of the beam framework can be analyzed by means of an eigenvalue approach. This method is particularly efficient for homogeneous systems without external loads,

as it avoids the numerical integration of the equations of motion and directly provides the natural frequencies and vibration modes of the structure. The starting point is the homogeneous differential-algebraic equation (DAE)

$$M_e \ddot{x}(t) + S_e x(t) = 0,$$

where

$$x(t) = [w(t) \ \mu(t)]^T$$

contains the displacement degrees of freedom and the Lagrange multipliers associated with the constraints. The matrices  $M_e$  and  $S_e$  denote the extended mass and stiffness matrices introduced in the previous sections.

### 2.3.2 Standing-Wave Ansatz

To determine the free vibration modes, solutions of the form

$$x(t) = \sigma(t)x_0$$

are considered, where  $x_0$  is a non-zero constant vector and  $\sigma(t)$  is a scalar amplitude function.

This assumption represents a standing wave: the spatial deformation pattern remains fixed while its amplitude oscillates in time. Substituting the ansatz into the homogeneous DAE yields

$$\ddot{\sigma}(t)M_e x_0 + \sigma(t)S_e x_0 = 0.$$

Since  $S_e$  is invertible, the matrix

$$A = S_e^{-1}M_e$$

can be introduced. The previous equation then leads to the eigenvalue problem

$$Ax_k = \lambda_k x_k.$$

Each eigenvector

$$x_k = [w_k \ \mu_k]^T$$

defines a vibration mode, while the corresponding eigenvalue  $\lambda_k$  determines the natural frequency

$$\omega_k = \frac{1}{\sqrt{\lambda_k}}.$$

### 2.3.3 Modal Orthogonality

The eigenvectors satisfy the orthogonality relation

$$w_j^T M w_k = 0, \quad j \neq k.$$

This property is the discrete analogue of the orthogonality of continuous beam eigenfunctions. It allows the vibration modes to be treated independently and forms the basis for the modal decomposition of arbitrary free vibrations.

### 2.3.4 General Solution

The complete solution of the homogeneous DAE can be expressed as a linear combination of all vibration modes:

$$x(t) = \sum_{k=1}^{N-K} \left( \alpha_k \cos(\omega_k t) + \frac{\beta_k}{\omega_k} \sin(\omega_k t) \right) x_k.$$

The coefficients  $\alpha_k$  and  $\beta_k$  describe the contribution of the  $k$ -th mode and are determined from the initial displacement and velocity fields.

Using modal orthogonality, the coefficients are obtained as

$$\alpha_k = \frac{w_k^T M w(0)}{w_k^T M w_k},$$

and

$$\beta_k = \frac{w_k^T M \dot{w}(0)}{w_k^T M w_k}.$$

Consequently, any admissible free vibration can be represented as a superposition of standing waves.

### 2.3.5 Analytical Implementation

To verify the correctness of the finite element implementation, the numerical results are compared against the analytical solution of a cantilever beam undergoing free vibration. For an Euler–Bernoulli cantilever beam of length  $L$ , the mode shapes and natural frequencies are known in closed form.

The analytical eigenvalues are determined from the characteristic equation

$$\cosh(\kappa_j L) \cos(\kappa_j L) + 1 = 0,$$

whose roots  $\kappa_j L$  define the wave numbers. The corresponding natural frequencies are given by

$$\omega_j = \kappa_j^2 \sqrt{\frac{EI}{\mu}},$$

where  $E$  is the Young's modulus,  $I$  the second moment of area, and  $\mu$  the mass per unit length.

The associated mode shapes are expressed as

$$w_j(x) = A[(\cosh(\kappa_j x) - \cos(\kappa_j x)) - \frac{\cosh(x_j) + \cos(x_j)}{\sinh(x_j) + \sin(x_j)} (\sinh(\kappa_j x - \sin(\kappa_j x)))]$$

where  $x_j = L\kappa_j$ . The factor  $A$  is chosen as  $A = L^{-\frac{1}{2}}$ .

In the implementation, the following steps are carried out:

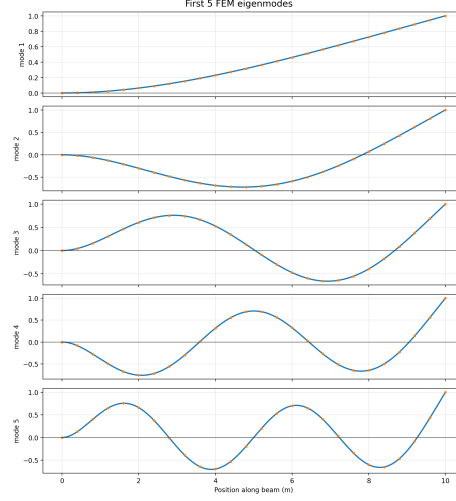


Figure 6: First five FEM eigenmodes

1. Compute the first few analytical roots  $\kappa_j L$  numerically.
2. Evaluate the corresponding analytical natural frequencies  $\omega_j$ .
3. Assemble the finite element mass matrix  $M$  and stiffness matrix  $S$  for the cantilever beam.
4. Solve the discrete eigenvalue problem

$$Sw_j = \omega_j^2 M w_j.$$

5. Compare the numerical eigenfrequencies and mode shapes with the analytical reference solution.

This comparison serves as a validation step for the code. Agreement between analytical and numerical frequencies, as well as qualitative similarity of the mode shapes, confirms the correctness of the finite element formulation and its implementation.

### 2.3.6 Results and Discussion

The eigenvalue analysis was carried out for the cantilever beam using the finite element framework developed in this project. The first five eigenmodes and their corresponding eigenfrequencies were computed and compared with the analytical solution.

The first five FEM eigenmodes were visualized and shown in Fig.6.

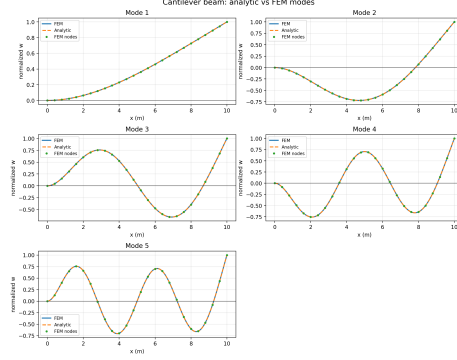


Figure 7: Comparison between eigenmodes and analytical modes

To compare eigenvalue method and analytical method, the first five analytical mode shapes with the corresponding FEM eigenmodes were also visualized and compared, as shown in Fig. 7.

An excellent agreement between the numerical and analytical solutions can be observed for all investigated modes. The FEM nodal values lie directly on the analytical curves, and no visible discrepancies can be identified in the plots. This confirms that the finite element discretization accurately reproduces the vibration modes of the beam and that the eigenvalue formulation has been implemented correctly.

Furthermore, the expected increase in modal complexity can clearly be observed. While the first mode exhibits a smooth monotonic deformation shape, higher modes contain additional nodal points and oscillations along the beam length. This behavior is consistent with the theoretical properties of cantilever beam vibration.

The relative errors of the first five eigenfrequencies are shown in Table2.

| Mode | $\omega_{FEM}(\text{rad/s})$ | $\omega_{analytic}(\text{rad/s})$ | relative error |
|------|------------------------------|-----------------------------------|----------------|
| 1    | 1.78e2                       | 1.78e2                            | 2.12e-8        |
| 2    | 1.12e3                       | 1.12e3                            | 8.61e-7        |
| 3    | 3.12e3                       | 3.12e3                            | 6.73e-6        |
| 4    | 6.12e3                       | 6.12e3                            | 2.57e-5        |
| 5    | 1.01e4                       | 1.01e4                            | 7.00e-5        |

Table 2: Relative error between FEM method and analytical solutions

The numerical frequencies agree extremely well with the analytical reference values. Although the error increases with the mode number, as shown in Fig.8, the growth remains very small. This trend is expected because higher vibration modes possess shorter wavelengths and therefore require a finer spatial representation.

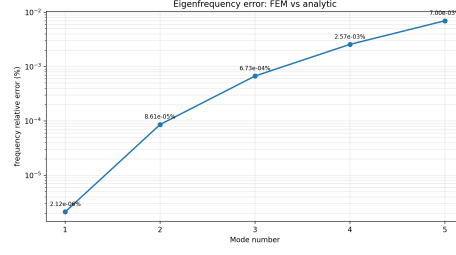


Figure 8: Frequency errors of eigenvalue method

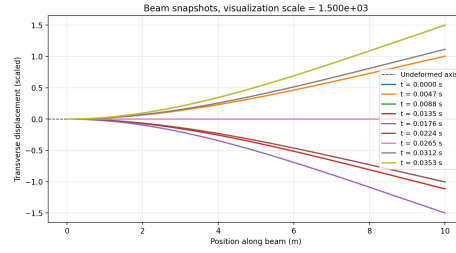


Figure 9: Snapshots of the standing wave

Fig.9 illustrates several snapshots of the first vibration mode reconstructed using the eigenvalue solution.

The deformation pattern retains its spatial shape throughout the motion, while only the amplitude changes with time. At specific times the amplitude vanishes and the beam passes through its undeformed configuration. Afterwards the deformation reappears with opposite sign.

This behavior is characteristic of a standing wave and confirms the theoretical assumption:

$$w(x, t) = \sigma(t)\phi(x)$$

which forms the basis of the eigenvalue method.

The results demonstrate that the eigenvalue method provides an efficient and highly accurate approach for analyzing the free vibration of beam structures.

Compared with the Newmark method, no time-stepping procedure is required to determine the natural frequencies and mode shapes. Once the eigenvalue problem has been solved, the vibration response can be reconstructed directly through modal superposition.

The close agreement between analytical and numerical results confirms the correctness of the finite element implementation, the assembly of the mass and

stiffness matrices, and the treatment of the boundary conditions. The computed eigenmodes form a suitable basis for representing arbitrary free vibrations of the beam.

### 3 Extension to a Beam Framework

As the final stage of the project, the finite element model is extended from a single beam to a two-dimensional framework composed of multiple interconnected beam elements. Building upon the methods developed in the previous sections, the framework is constructed by assembling the element matrices into a global system and enforcing the structural constraints through a unified constraint matrix. The resulting model is then analyzed using the same computational procedures as for the single-beam case, including static equilibrium analysis, Newmark time integration, and eigenvalue-based free vibration analysis.

#### 3.1 Geometry of the Framework

##### 3.1.1 Framework geometry

The proposed framework consists of five Bernoulli beam elements and four connection nodes, as shown in Fig.???. The framework consists of one fixed support at node  $p_1$  and one partially constrained support at node  $p_4$ . The support at  $p_1$  prevents both translational and rotational motion, while the support at  $p_4$  allows motion only in the global vertical direction by constraining the horizontal displacement. The remaining nodes ( $p_2$  and  $p_3$ ) are internal connection points where multiple beam elements are rigidly connected.

##### 3.1.2 Degrees of freedom

Unlike the single-beam model, where only transverse bending is considered, each beam in the framework is allowed to deform both axially and transversely. Therefore, every node possesses three degrees of freedom:

- axial displacement  $v$ ;
- transverse displacement  $w$ ;
- rotation  $\theta$ .

To simplify the computations and focus on the structure of the whole system, each beam is regarded as only one element. Thus, for each beam  $e$ , the DOF vector is given as:

$$u_e = [v_0, w_0, \theta_0, v_1, w_1, \theta_1]^T$$

Consequently, each beam element contributes six degrees of freedom to the global system.

### 3.1.3 Global assembly

In the following studies, each beam is first formulated in its own local coordinate system. The element mass and stiffness matrices are then assembled into the global matrices by:

$$M = \text{diag}(M_1, \dots, M_5)$$

$$S = \text{diag}(S_1, \dots, S_5)$$

At this stage the beam elements are still independent. The structural connectivity is introduced through the constraint equations described in the next section.

## 3.2 Constraints formulation

After assembling the element matrices into the global system, the connectivity between adjacent beam elements is automatically enforced through the shared nodal degrees of freedom. Therefore, unlike the extended matrix formulation presented in the lecture notes, no additional compatibility equations are required for the internal joints. The constraint matrix is only used to impose the prescribed support conditions of the framework.

We introduced three kinds of constraints in the proposed framework.

### 3.2.1 Fixed support

*Beam1* and *Beam2* are connected at node  $p_1$  by a fixed bearing, which suppresses both translational and rotational motion. The following has to be satisfied:

$$v_{1,0} = v_{2,0} = 0, w_{1,0} = w_{2,0} = 0, \theta_{1,0} = \theta_{2,0} = 0.$$

These constraints prevent rigid-body translation and rotation of the framework and provide a fixed reference for the structural deformation.

### 3.2.2 Partially constrained support

Node  $p_4$  is modeled as a partially constrained (sliding) support. The node is only allowed to move in the global vertical direction while its horizontal displacement is constrained. The supporting condition is therefore

$$v_{4,1} = 0$$

Since the element formulation is assembled directly in the global coordinate system, no coordinate transformation is required when applying this boundary condition.

### 3.2.3 Compatibility conditions

Unlike the extended matrix formulation presented in the lecture notes, the present implementation adopts the standard finite element assembly procedure. Beam elements connected to the same node share the same global degrees of freedom  $([v, w, \theta]^T)$ . As a result, displacement compatibility and rotational continuity are automatically satisfied during the assembly process, and no additional compatibility equations are required.

For example, node  $p_2$  is shared by Beam 1, Beam 3, and Beam 5. These three elements reference the same global degrees of freedom,

$$(v_2, w_2, \theta_2),$$

during the assembly of the global stiffness and mass matrices. Consequently, all connected beam elements always possess identical translations and rotations at the joint, ensuring structural continuity without introducing additional constraint equations. This approach corresponds to the standard finite element assembly procedure for two-dimensional frame structures.

### 3.2.4 Constraint matrix

The prescribed support conditions are assembled into a global constraint matrix  $C$ . Since the continuity between adjacent beam elements is automatically enforced through the shared nodal degrees of freedom, the constraint matrix is only used to impose the boundary conditions of the framework. The complete set of constraints is written compactly as

$$C^T u = 0,$$

where  $u$  denotes the global displacement vector of the framework.

For the proposed framework, only four constraints are required. The fixed support at node  $p_1$  constrains the horizontal displacement, vertical displacement, and rotation,

$$v_1 = 0, \quad w_1 = 0, \quad \theta_1 = 0,$$

while the sliding support at node  $p_4$  constrains only the horizontal displacement,

$$v_4 = 0.$$

Since the framework consists of four nodes with three global degrees of freedom per node, the global displacement vector has twelve components,

$$u \in \mathbb{R}^{12},$$

and the corresponding constraint matrix is

$$C \in \mathbb{R}^{12 \times 4}.$$

The resulting constrained system is subsequently employed in the static analysis, Newmark time integration, and eigenvalue analysis without requiring any modification of the underlying numerical algorithms.

### 3.3 Static simulation of the framework

To verify the correctness of the assembled framework model, a static analysis is first performed. A concentrated vertical load is applied at the selected loading node (node  $p_3$ ), while the support conditions introduced in Section 5.2 are enforced throughout the solution process. Since the compatibility conditions between connected beam elements are implemented by shared degrees of freedom, the assembled framework behaves as a continuous structure without requiring additional coupling equations at the internal joints.

The static equilibrium of the constrained framework is obtained by solving

$$\begin{bmatrix} S & C \\ C^T & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda \end{bmatrix} = \begin{bmatrix} f \\ a \end{bmatrix}$$

where  $S$  denotes the global stiffness matrix,  $u$  is the global displacement vector,  $C$  represents the support constraint matrix, and  $\lambda$  contains the corresponding Lagrange multipliers. In the present implementation, the explicit constraint matrix is only required for the external support conditions, whereas the compatibility conditions at the internal joints are satisfied implicitly through the shared nodal degrees of freedom.

Fig.10 compares the undeformed and deformed configurations of the framework under the applied load. For visualization purposes, the deformation is magnified by a constant scale factor. The smooth deformation of each beam is reconstructed using the Hermite shape functions employed in the finite element formulation. Although each beam is represented by a single finite element, the displacement field inside each element is evaluated at multiple sampling points during post-processing, producing a continuous representation of the deformed geometry.

To further illustrate the structural response, the displacement magnitude is evaluated along each beam as

$$|\mathbf{u}(x)| = \sqrt{u_x(x)^2 + u_y(x)^2},$$

and visualized using a displacement contour, as shown in Fig.11. The contour is obtained by interpolating the nodal displacements with the element shape functions.

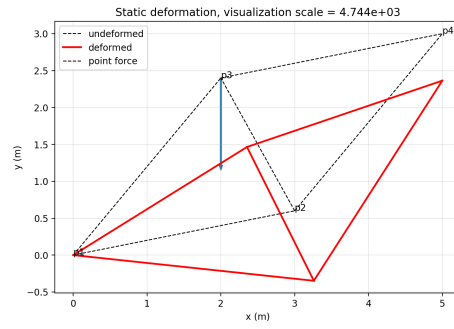


Figure 10: Comparison between undeformed and deformed static framework

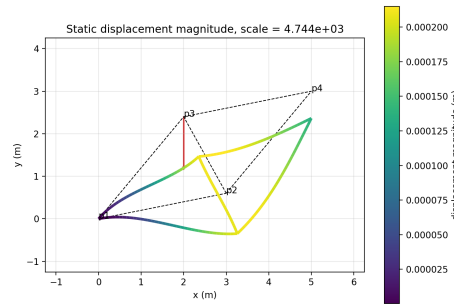


Figure 11: Contour under point load

The results show that the deformation is continuous across all beam connections, confirming that the shared-degree-of-freedom assembly correctly enforces displacement compatibility. As expected, the fixed support at  $p_1$  remains immobile, while the partially constrained support at  $p_4$  permits only vertical motion. The maximum displacement occurs near the loaded region and gradually decreases towards the supports, indicating that the framework responds consistently with the prescribed boundary conditions and the expected bending behaviour.

### 3.4 Dynamic simulation of the framework

The transient response of the framework is investigated using the Newmark time integration method developed in the previous section. The global mass matrix, stiffness matrix, and support constraints assembled for the framework are employed directly, while the compatibility conditions between adjacent beam elements are inherently satisfied through the shared nodal degrees of freedom.

The dynamic equilibrium of the framework is governed by

$$M\ddot{u} + Ku = f(t),$$

subject to the prescribed support constraints. In the present study, the average acceleration Newmark scheme is adopted with

$$\beta = \frac{1}{4}, \quad \gamma = \frac{1}{2},$$

which is unconditionally stable for linear elastic systems.

To excite the structural response, a transverse force is applied at node  $p_3$ . In this study, two loading scenarios are considered: the force  $q_0$  is treated both as an initial instantaneous load applied at  $t = 0$ , and as a constant force maintained throughout the simulation. These loading conditions allow the investigation of both free vibration initiated by an impulse and forced vibration under sustained excitation, ensuring that multiple vibration modes are activated and the dynamic behaviour of the system can be comprehensively observed.

At each time step, the Newmark algorithm computes the updated nodal displacements, velocities, and accelerations. The displacement field inside every beam is reconstructed using the same interpolation functions as in the static analysis, allowing a continuous deformation of the framework to be visualized throughout the simulation.

#### 3.4.1 Instantaneous force response

In the second loading scenario, the same load magnitude is applied only at the initial instant and removed immediately afterwards. The structure therefore

undergoes free vibration without any sustained external excitation.

The corresponding deformation sequence is presented by snapshots in Fig.12. The displacement history of node  $p_3$  exhibits a nearly periodic response composed of multiple vibration modes excited by the initial impulse, as shown in Fig.13.

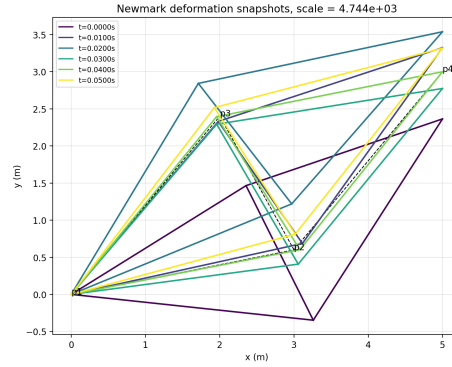


Figure 12: Snapshots of the framework under instantaneous force

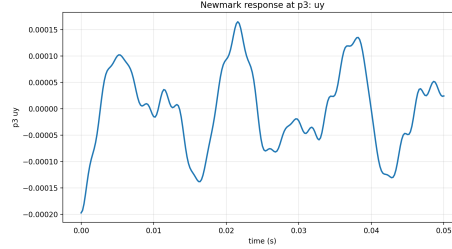


Figure 13: Displacement of node  $p_3$  under instantaneous force

The total mechanical energy remains nearly constant throughout the simulation, as shown in Fig.14. indicating that the Newmark implementation conserves energy well for the undamped system. Small fluctuations are attributed to numerical discretization errors.

### 3.4.2 Constant force condition

We also studied the case when a constant transverse force  $q_0$  is continuously applied throughout the simulation,

$$f(t) = q_0.$$

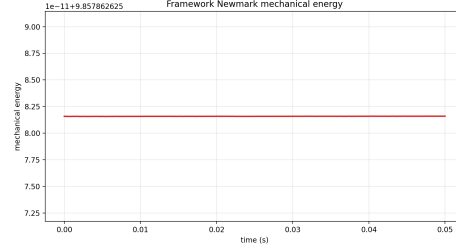


Figure 14: Energy wave of the framework under instantaneous force

Fig.15 illustrates the deformation of the framework at several representative time instants. Starting from the undeformed configuration, the structure accelerates under the external load and gradually approaches a dynamically oscillating equilibrium state.

Since the external force continuously performs work on the structure, the total mechanical energy increases during the initial stage before oscillating around a higher energy level, as shown in Fig.16.

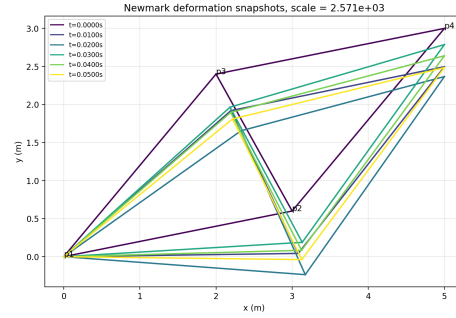


Figure 15: Snapshots of the framework

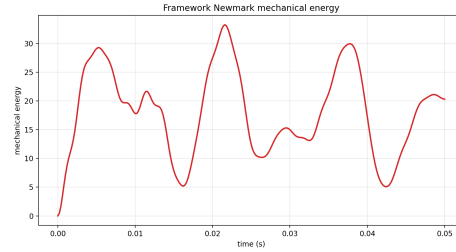


Figure 16: Energy wave of the framework

The displacement history of node  $p_3$  is shown in Fig.17. The response ex-

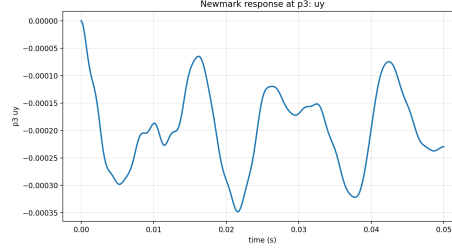


Figure 17: Displacement of node  $p_3$  under constant force

hibits oscillatory behaviour superimposed on a non-zero mean displacement, reflecting the static deformation introduced by the constant external force together with the dynamic vibration. The deformation remains continuous throughout the framework, confirming the correctness of the shared-degree-of-freedom assembly and the stability of the Newmark integration scheme.

### 3.4.3 Comparison of the two loading cases

The two loading scenarios produce significantly different dynamic characteristics despite having the same initial load magnitude. Under the constant force, the framework experiences both a static deformation and a superimposed dynamic response. Continuous external work results in a higher overall energy level and a displacement oscillating around a shifted equilibrium position.

In contrast, the instantaneous load excites the natural vibration modes of the framework without further energy input after the initial excitation. Consequently, the displacement oscillates about the undeformed configuration and the total mechanical energy remains approximately conserved. These two cases demonstrate the capability of the proposed framework to simulate both forced and free vibration using the same Newmark integration procedure.

## 3.5 Eigenvalue analysis of the framework

To investigate the intrinsic dynamic characteristics of the beam framework, an eigenvalue analysis is performed based on the assembled finite element model. In the absence of external excitation, the free vibration of the structure is governed by

$$M\ddot{u} + Ku = 0,$$

together with the prescribed support constraints. Since the compatibility conditions between connected beam elements are enforced through shared nodal degrees of freedom during the assembly process, the eigenvalue analysis is performed directly on the assembled global matrices.

Assuming harmonic motion,

$$u(t) = \phi e^{i\omega t},$$

the governing equation is transformed into the generalized eigenvalue problem

$$K\phi = \omega^2 M\phi,$$

where  $\omega$  denotes the natural angular frequency and  $\phi$  is the corresponding mode shape. Solving this system yields the natural frequencies and vibration modes of the framework.

### 3.5.1 Natural frequencies

Table 3 lists the first several natural frequencies obtained from the finite element model. As expected, the fundamental mode possesses the lowest frequency and dominates the global deformation of the structure, whereas higher-order modes correspond to increasingly complex deformation patterns involving multiple beam members.

The computed frequencies provide the characteristic dynamic properties of the framework and serve as a reference for interpreting the transient responses obtained from the Newmark simulations.

| Mode | $\omega(rad/s)$ | Frequencies(Hz) |
|------|-----------------|-----------------|
| 1    | 4.32e2          | 68.77           |
| 2    | 6.90e2          | 109.84          |
| 3    | 7.51e2          | 119.51          |
| 4    | 1.22e3          | 194.77          |
| 5    | 1.54e3          | 245.65          |
| 6    | 2.55e3          | 405.24          |

Table 3: Frequencies of the framework

### 3.5.2 Mode shapes

The corresponding mode shapes are illustrated in Fig.18. Similar to the static and dynamic visualizations, the deformation within each beam is reconstructed using Hermite interpolation, producing smooth mode shapes despite each beam being represented by a single finite element.

The corresponding mode shapes are illustrated in Figure X. Similar to the static and dynamic visualizations, the deformation within each beam is reconstructed using Hermite interpolation, producing smooth mode shapes despite each beam being represented by a single finite element.

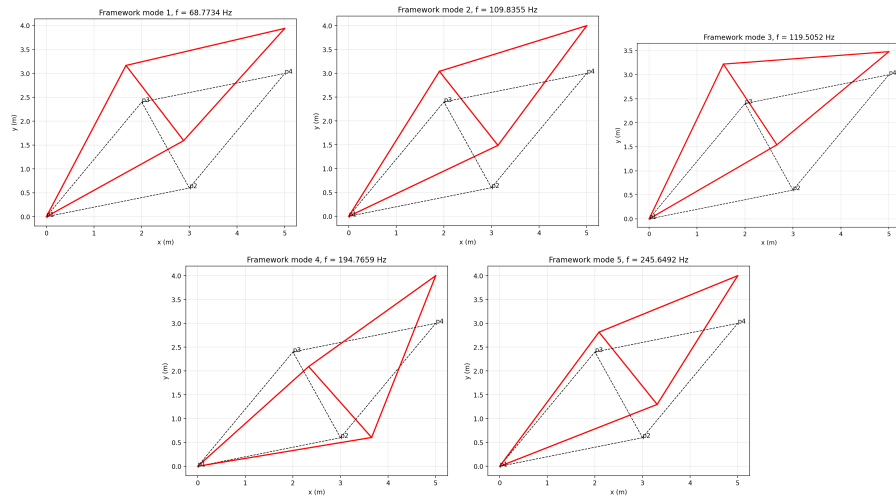


Figure 18: First five mode shapes of the framework